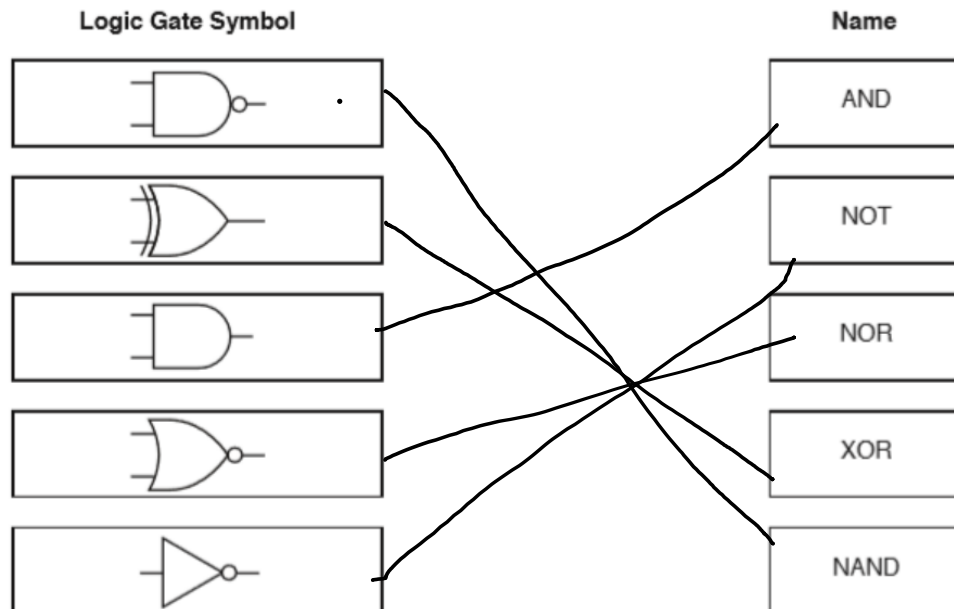


IN0013 week 2 exercise

Logic gates

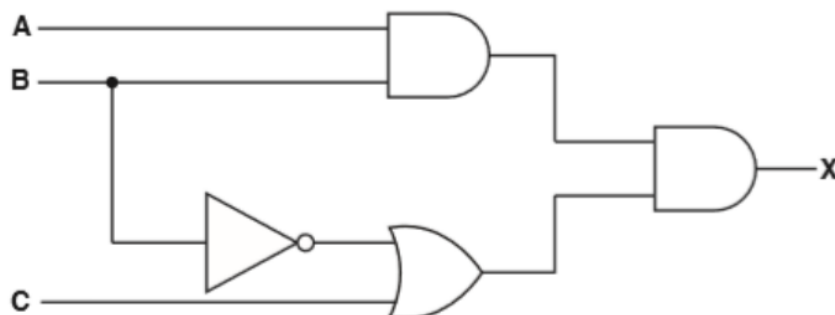
- The diagram below shows 5 logic gates and 5 names. Draw a line matching the logic gates with the respective names



- Complete the truth table for the NOR gate

A	B	X
1	1	0
1	0	0
0	1	0
0	0	1

- Write a logic statement that corresponds to the following logic circuit



$$X = (A \text{ AND } B) \text{ AND } (\text{NOT } B \text{ OR } C)$$

Answer: (A and B) and (not B or C)

4. One way to think of the basic logic gate types (all but the XOR and XNOR gates) is to consider what single input state guarantees a certain output state. For example, we could describe the function of an OR gate as such:

Any high input guarantees a high output.

Identify what type of gate is represented by each of the following phrases:

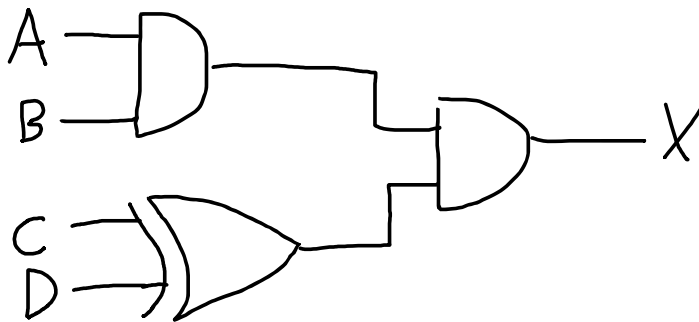
- Any high input guarantees a low output: **NOR**
 - Any low input guarantees a high output: **NAND**
 - Any low input guarantees a low output: **AND**
5. Design a logic circuit to model the requirement for a gym membership. To become a member of this prestigious gym ($X = 1$), the 4 criteria requirements are listed below

Parameter	Description of parameter	Binary value	Condition
A	Over 18	1	True
		0	False
B	Recommended	1	True
		0	False
C	Full-time	1	True
		0	False
D	Retired	1	True
		0	False

Membership is approved ($X = 1$) if the person:

- is over the age of 18 and has been recommended by a pre-existing member and
- either is working full-time or is retired, but not both.

Draw a logic circuit to represent the membership requirement and draw the truth table showing the logic results. $X = (A \text{ and } B) \text{ and } (C \text{ xor } D)$



A	B	C	D	A AND B	C XOR D	(A AND B) AND (C XOR D)
0	0	0	0	0	0	0
0	0	0	1	0	1	0
0	0	1	0	0	1	0
0	0	1	1	0	0	0
0	1	0	0	0	0	0
0	1	0	1	0	1	0
0	1	1	0	0	1	0
0	1	1	1	0	0	0
1	0	0	0	0	0	0
1	0	0	1	0	1	0
1	0	1	0	0	1	0
1	1	0	0	1	0	0
1	1	0	0	1	0	0
1	1	0	1	1	1	1
1	1	1	0	1	1	1
1	1	1	1	1	0	0

Binary

- Write the following decimal number as a binary number
 - 4 = 0100
 - 11 = 1011
 - 16 = 0001 0000
 - 19 = 0001 0011
 - 59 = 0011 1011
- Write the following binary number as a decimal number
 - 11 = 3
 - 110 = 6
 - 1100 = $2^3 + 2^2 + 0 + 0 = 8 + 4 = 12$
 - 1101 = 11

e. $101111 = 15 + 32 = 47$

3. Add these binary numbers

a. $11 + 11 = 110$

b. $1100 + 0101 = 0001\ 0001$

c. $1000 + 0011 = 1011$

d. $10101 + 01111 = 16 + 1 + 4 = 21 + 15 = 36$

4. Two complements – use two complements to compute these binary operations

a. $110 - 101 =$

$$\begin{array}{r} 001 \\ 011 \\ - 101 \\ \hline 110 \\ + 011 \\ \hline 1001 \end{array}$$

b. $1001 - 0111 =$

$$\begin{array}{r} 0010 \\ 1001 \\ - 0111 \\ \hline 1001 \\ + 1001 \\ \hline 0010 \end{array}$$

c. $1000 - 0011 = 1000 + (1100) = 1000 + (1101) = 0101$

d. $10101 - 01111 =$

$$\begin{array}{r} 0110 \\ 10101 \\ - 01111 \\ \hline 10101 \\ + 10001 \\ \hline 100110 \end{array}$$

5. ASCII numbers – transform these letters into ascii or the other way around

a. computer = $99\ 111\ 109\ 112\ 117\ 116\ 101\ 114$

b. fundamental = $102\ 117\ 110\ 100\ 97\ 109\ 101\ 110\ 116\ 97\ 108$

c. $80\ 121\ 116\ 104\ 111\ 110 = \text{Python}$